SteemBiota

Immutable Evolution

White Paper

A Decentralised Life Simulation on the Steem Blockchain

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Abstract

SteemBiota is a fully decentralised life simulation built on the Steem blockchain. Every creature is defined by a compact, immutable ten-gene genome that determines its visual appearance, lifespan, fertility window, and mutation tendency. From the moment a creature is published as a Steem post, its entire existence — feeding, play, walking, breeding, ownership transfer, and eventual fossilisation — is permanently recorded on-chain as a series of blockchain replies.

The project demonstrates that a rich, interactive game ecosystem can be built entirely on a public blockchain without any backend server, centralised database, or trusted operator. All game logic runs client-side in the browser. The blockchain is both the data store and the source of truth. Creatures are provably authentic, family trees are verifiably traceable, and every transaction is auditable by any participant.

This white paper describes the architecture, genome system, lifecycle model, breeding mechanics, provenance system, ownership transfer protocol, and community incentive layer that together constitute the SteemBiota ecosystem.

1. Introduction

Blockchain technology is most commonly associated with financial instruments — tokens, smart contracts, and decentralised exchanges. SteemBiota explores a different application: using a public blockchain as the permanent substrate for a living digital ecosystem.

The central premise is simple. If a blockchain can record an immutable ledger of financial transactions, it can equally record an immutable ledger of biological events: births, feedings, breedings, and deaths. The result is a fossil record that no single party can alter or delete.

SteemBiota is built on the Steem blockchain because Steem is designed as a social content platform. Every piece of content — posts, replies, metadata — is stored on-chain with a permanent timestamp and author signature. This makes Steem an unusually natural fit for a life simulation: creatures are posts, interactions are replies, and the blockchain's existing social graph infrastructure handles content discovery and display for free.

1.1 Design Philosophy

- **Immutability:** Once a genome is published, it cannot be changed. A creature's DNA is permanent.
- **Determinism:** The same genome always renders the same creature. The same two parents always produce the same child.
- **Client-side only:** All game logic runs in the browser. No servers, no databases, no trusted operators.
- **UTC time:** All timestamps use UTC to match the Steem blockchain clock, ensuring consistent lifecycle calculations across time zones.
- **Diversity enforcement:** The kinship system prevents same-bloodline farming and encourages genuine cross-community collaboration.
- **Provenance transparency:** Genome fingerprinting exposes copy-pasted creatures publicly without requiring any trusted authority.
- **Anti-dumping ownership:** Ownership transfers require the recipient's explicit on-chain acceptance. Creatures cannot be forced on unwilling users.
- **Opt-in breeding:** Creatures are closed to external breeding by default. Owners must explicitly grant permits, preventing unsolicited use of their creatures as parents.

2. Technology Stack

SteemBiota is a zero-dependency single-page application. It runs entirely in the browser with no build tools, no package manager, and no backend infrastructure.

Layer	Technology
Blockchain	Steem — a delegated proof-of-stake social blockchain with permanent on-chain post and reply storage
Signing	Steem Keychain — a browser extension that signs transactions with the user's Steem private key without exposing it to the application
UI Framework	Vue 3 (CDN) — reactive component framework loaded directly from CDN, no build step required
Routing	Vue Router 4 (CDN, hash mode) — client-side routing without server configuration
Hosting	GitHub Pages — static file hosting, no server logic

The codebase consists of four files: index.html (shell and styles), blockchain.js (all Steem API calls and publish functions), components.js (Vue component definitions), and app.js (genome logic, game systems, and route views). The entire application is readable as plain source code with no compilation step.

2.1 RPC Node Resilience

All Steem API calls are wrapped in a fallback chain across multiple public RPC nodes: api.steemit.com, api.justyy.com, steemd.steemworld.org, and api.steem.fans. If the primary node is unavailable or returns an error, the call is automatically retried against the next node in the chain. This provides resilience against individual node downtime without requiring any infrastructure management.

3. The Genome System

Every SteemBiota creature is defined by a compact genome: ten integer values that encode everything about the creature's identity, appearance, and lifecycle. The genome is generated once at the moment of creation, stored immutably on-chain, and never changes.

3.1 Gene Table

Gene	Description	Range	Role
GEN	Genus identifier — the species barrier	0 – 999	Taxonomy & colour palette
SX	Sex (0 = Male, 1 = Female)	0 – 1	Breeding compatibility
MOR	Morphology seed — body shape and tail style	0 – 9999	Visual body form
APP	Appendage seed — ear shape, paw shape, wing presence	0 – 9999	Visual appendages
ORN	Ornamentation seed — glow orbs, mane, body pattern	0 – 9999	Visual decoration
CLR	Colour hue offset applied on top of palette	0 – 359	Hue modulation
LIF	Base lifespan in real-world days	80 – 159	Lifecycle timing
FRT_START	Fertility window start (days since birth)	varies	Breeding eligibility
FRT_END	Fertility window end (days since birth)	varies	Breeding eligibility
MUT	Mutation tendency — affects offspring gene variation probability	0 – 2 (founders); 0 – 5 (offspring)	Hereditary variation

3.2 Deterministic Naming

Every creature has a two-part scientific name derived deterministically from its genome. The genus portion (e.g. Vyrex) is stable across all creatures sharing the same GEN value — it is computed from GEN alone, giving every genus a consistent identity. The species portion (e.g. Nymwhisper) is derived from ORN, CLR, and MUT, making it unique to the individual. The naming system draws from curated phoneme tables yielding approximately 5.3 million distinct combinations.

Names are used as Steem post titles, in blockchain post bodies, in the UI, and as post tags, making each creature's identity discoverable through Steem's existing tag and search infrastructure.

3.3 Genome Storage

The genome is stored in two locations within each Steem post, providing redundancy and flexibility for client applications:

- In json_metadata.steembiota.genome — structured JSON accessible via the Steem API without parsing post bodies.
- In a fenced genome code block in the post body — a human-readable fallback that survives even if the API's metadata parsing changes.

Any client with Steem API access can reconstruct a creature entirely from its post, independent of the SteemBiota application itself.

4. Visual Rendering

Each creature is rendered in two formats: a procedural canvas painting for the interactive web application, and a Unicode art representation stored permanently inside the Steem post body.

4.1 Canvas Rendering

The canvas renderer produces a 400×320 pixel procedural illustration of the creature on every page load. The same genome always produces the same base visual — colour palette, body proportions, appendage shapes, ornamentation, and facial features are all deterministic functions of the genome. Three sources of variation are layered on top per page load: a randomly chosen facing direction, a randomly chosen pose, and a live expression state driven by current game data.

4.1.1 Anatomy (painter's algorithm, back to front)

Ground shadow → energy ribbons → back legs (depth-dimmed) → tail → torso with gradient → chest marking → body pattern → front legs → neck → mane wisps → head with snout and nose → ears → eye → face expression overlay → dorsal wing/fin → glowing orb nodes → fertility aura

4.1.2 Poses

Pose	Description
Standing	Default upright side profile
Alert	Torso raised, head lifted high, tail swept straight up
Playful	Play-bow: front legs stretched forward and low, rear elevated, tail up
Sitting	Torso tilted rear-down, folded haunches resting on ground, tail wrapped under body
Sleeping	Body flat and low, head resting on ground, all legs tucked as flat pads, eye closed

4.1.3 Face Expressions

Expressions are computed from live game state (feed health and activity data) and re-evaluated whenever data refreshes. Pose overrides take highest priority.

Expression	Trigger	Visual Details
Sleepy	Sleeping pose	Closed arc eye, drooped brow, neutral mouth
Alert	Alert pose	Enlarged eye (×1.15), raised brow
Excited	Playful pose	Wide smile + tongue, star glints beside eye

Thriving	Feed health ≥ 80%	Big smile, raised brow, rosy cheek blush
Happy	Feed health ≥ 55%	Gentle smile, relaxed raised brow
Content	Health ≥ 30% or no data	Neutral mouth, flat brow
Hungry	0% < health < 30%	Slight frown, worried brow, pupil shifted down
Sad	Completely unfed (health = 0%)	Pronounced frown, teardrop below eye

4.1.4 Reaction Animation

Successful interactions trigger a reaction sequence: the creature cycles through four pose+expression pairs (Standing→Alert→Playful→Sitting, paired with Alert→Alert→Excited→Happy), repeated 2–3 times at 2–3 seconds per step. Any running animation is cancelled and restarted when a new interaction completes.

4.2 Unicode Rendering

A second renderer produces a text-art representation of the creature using Unicode block characters, box-drawing symbols, and emoji. This is embedded directly into the Steem post body, so the creature's visual form is stored permanently on-chain as plain text — readable in any Steem client and preserved forever in the blockchain's immutable record.

Art width scales with lifecycle stage: 14 characters at Baby, growing to 36 at Young Adult, then contracting to 30 at Elder and 24 at Fossil. Fertile creatures display sparkle characters in the header row.

5. Lifecycle

Creature age is measured in real-world days elapsed since the post's on-chain timestamp. Lifecycle stage is a deterministic function of age as a percentage of the creature's effective lifespan (base LIF plus any bonuses from feeding and walking).

Stage	Age % of Lifespan	Icon	Notes
Baby	0 – 4%	□	Smallest body scale (45%)
Toddler	5 – 11%	□	Expressions begin
Child	12 – 24%	□	Growing body scale
Teenager	25 – 39%	□	Approaching fertility
Young Adult	40 – 59%	□	Peak body scale (100%); fertility window active
Middle-Aged	60 – 79%	□	Fertility window may close
Elder	80 – 99%	□	Body scale reducing
Fossil	100%+	□	Immutable; no further interactions

Once a creature reaches Fossil stage, no further interactions are possible — it cannot be fed, played with, walked, or used in breeding. Its genome, unicode art, and full interaction history remain permanently on-chain as a complete biological record.

5.1 Phantom Creatures

If a Steem post is deleted via the delete_comment operation, the blockchain API returns it with an empty author field. SteemBiota detects this tombstone state as a Phantom — distinct from a natural Fossil. Phantoms display a memorial screen and cannot participate in any interactions. If any ancestor in a breeding chain is phantom, the breeding attempt is blocked entirely, preventing genealogy evasion attacks.

6. Activities

Three activity types are available on every living, non-fossil creature. All are published as blockchain replies with structured json_metadata, creating a permanent, auditable record of every interaction. Anti-spam rules are enforced client-side and verified by any compliant client reading the same chain data.

6.1 Feeding

Feeding improves a creature's health, which affects its canvas expression, lifespan, and fertility window. Three food types are available:

Food	Lifespan Bonus	Fertility Boost
Nectar □	+1 day per feed	None
Fruit □	+0.5 days per feed	+10% per community feed
Crystal □	None	+5% per community feed

Feed rules enforced on-chain:

- Each feeder is counted at most once per UTC calendar day (anti-spam).
- Total feeds per creature are capped at 20 across its lifetime.
- Owner feeds count 3× toward the health score; community feeds count 1×.
- Maximum lifespan bonus from feeding: +20% of base LIF.
- Maximum fertility boost from community feeding: +25% of the base window width.

6.2 Play

Playing improves a creature's mood, which widens its effective fertility window and boosts its canvas expression. Owner plays count 2×; community plays count 1×. The lifetime cap is 15 play events. At maximum mood, the fertility window is extended by up to +10 days on each side. Play activity also adds up to +25% to the effective health score used to determine face expression.

6.3 Walking

Walking builds a creature's vitality, which extends its effective lifespan. Owner walks count 2×; community walks count 1×. The lifetime cap is 15 walk events. At maximum vitality, the creature gains up to +10 additional days of life beyond its base LIF. Vitality is displayed as a teal badge on the creature page.

7. Breeding

Breeding is the core vector of genetic evolution in SteemBiota. Two compatible creatures produce a deterministic offspring whose genome is a seeded blend of both parents' genes, with possible mutations. The offspring is published as a new Steem post, creating an immutable on-chain record of the lineage.

7.1 Compatibility Rules

All five conditions must hold simultaneously for a breed to be permitted:

- Same GEN value (same genus — the species barrier).
- Opposite SX values (one Male, one Female).
- Both creatures are within their effective fertility window at the moment of breeding.
- Neither creature is a close relative of the other (see Kinship Rules).
- The breeding user holds a valid breed permit for both creatures (see Breed Permits).

7.2 Gene Inheritance

The child genome is computed using a seeded PRNG (mulberry32) keyed on a deterministic breed seed derived from both parent genomes. This guarantees that the same two parents always produce the same child — the result is verifiable and reproducible by any client.

Each gene is inherited from one parent chosen at random (50/50 probability). The gene is then potentially mutated: mutation probability per gene = 1% × (1 + MUT_A + MUT_B), where MUT_A and MUT_B are the mutation tendency genes of the respective parents. Higher MUT parents therefore produce more genetically varied offspring.

7.3 Speciation

There is a 0.5% probability per breeding event that the child's GEN gene mutates to an entirely new value, creating a new genus. Speciated offspring are incompatible with their parents' genus for future breeding, enforcing genuine taxonomic divergence. Speciation events are flagged on the offspring post and earn additional XP for the breeding user.

7.4 Kinship Rules

SteemBiota performs a client-side BFS traversal of the blockchain ancestry graph before permitting any breed. The check walks up to 12 generations. A breed is blocked if either creature is:

- A parent, grandparent, or any ancestor of the other.
- A sibling (full or half — any creature sharing at least one parent).
- A child, grandchild, or any descendant of the other.
- An aunt, uncle (parent's sibling) of the other.
- A niece, nephew, or any of their descendants.

When a breed is blocked, the UI displays the specific relationship that prevents it. If any ancestor in the chain is a phantom (deleted post), the check throws an error rather than silently passing, preventing inbreeding-check evasion through post deletion.

8. Breed Permits & Ownership Transfer

8.1 Breed Permits

Creatures are closed to external breeding by default. The effective owner must explicitly grant a named breed permit to allow another user to use their creature as a breeding parent.

Permits are published as breed_permit reply posts on the creature's post. Each permit names a specific grantee and an optional expiry period in days (0 = permanent). A breed_revoke reply cancels an existing permit. The latest action per grantee wins; expired permits are treated as revoked. Permits issued before a completed ownership transfer are automatically voided — the new owner starts with a clean slate.

The effective owner always holds implicit breed permission on their own creatures, requiring no explicit permit.

8.2 Ownership Transfer

Creature ownership can be transferred between users via a two-sided on-chain handshake that prevents unwanted transfers — a creature cannot be forced on an unwilling recipient.

Transfer Protocol

- **Step 1 — Offer:** The current effective owner publishes a transfer_offer reply naming the recipient. Only one pending offer may exist at a time; a new offer replaces the previous one.
- **Step 2 — Accept:** The named recipient publishes a transfer_accept reply referencing the exact offer permlink. This is the moment effective ownership changes.
- **Cancellation:** The current owner may publish a transfer_cancel reply at any time before acceptance to withdraw the offer.

Effective Owner Resolution

The original post.author never changes on-chain. SteemBiota derives the effective owner by walking the reply history via parseOwnershipChain() at read time. The effective owner governs breed permit management, transfer authority, profile page appearance, and implicit breed permission.

9. Provenance & Copy Detection

Because the Steem blockchain is public, genome data is readable by anyone. SteemBiota includes a provenance system that detects and publicly surfaces copy-pasted creatures without requiring any centralised authority.

9.1 Genome Fingerprint

Every creature carries a stable fingerprint: a pipe-delimited string of all ten gene values (GEN|SX|MOR|APP|ORN|CLR|LIF|FRT_START|FRT_END|MUT). Two posts sharing an identical fingerprint have identical genomes regardless of author, declared type, or parent links.

9.2 Timestamp Priority

When two posts share an identical fingerprint, the post with the earlier on-chain timestamp is the original. Any later post is flagged as a Duplicate, regardless of whether the copier also included the original parent links.

9.3 Provenance Badges

Every creature card and creature page display a provenance badge. Priority order (highest first):

Badge	Meaning
Phantom	Post was tombstoned on-chain — creature no longer exists
⚠ **Duplicate**	Identical genome exists in an earlier post — this is a copy
⚠ **No Parents**	Claims to be a bred offspring but contains no parent links in metadata
⚠ **Unverified Origin**	Posted as founder but genome has ≥ 3 simultaneously maxed traits (statistically implausible)
⚡ **Speciation**	Legitimately bred offspring that created a new genus
Bred	Legitimately bred offspring with verified parent links
Origin Creature	Legitimate randomly-generated founder

10. Notifications

The Notifications page aggregates all on-chain events relevant to the logged-in user. Because Steem has no server-side notification API, all event discovery is performed entirely client-side.

Events tracked:

- Feed — someone fed one of your creatures.
- Play — someone played with one of your creatures.
- Walk — someone walked one of your creatures.
- Birth — someone bred an offspring from one of your creatures.
- Breed — someone used your creature's lineage to breed a new creature.
- Transfer offer — someone is offering you ownership of a creature.

Incoming transfer offers are highlighted in a dedicated banner with a one-click Accept button. A notification badge on the navigation bar shows the count of pending transfer offers, refreshed on login and polled every five minutes.

11. User Levels & XP

Every on-chain action earns XP for the acting user. XP is computed client-side from blockchain history and displayed on each user's profile page and the global leaderboard. The XP system incentivises ecological contribution — not just creature creation, but active community participation through feeding, diverse genus exploration, and the coordination-intensive act of successful breeding.

Action	XP
Publish a founder creature	**100**
Publish an offspring	**500**
Each unique genus contributed (distinct GEN values across own creatures)	**25**
Each speciation event in own offspring	**75**
Feed a creature	**10**

Breeding an offspring is worth 5× more XP than publishing a founder because it requires two compatible, fertile, unrelated creatures to exist simultaneously — a significantly higher coordination barrier that represents genuine ecological investment.

11.1 Rank Thresholds

Rank	Minimum XP	Icon
Wanderer	0	□
Naturalist	100	□
Cultivator	300	□
Breeder	700	□
Ecologist	1,500	□
Evolutionist	3,000	□
Progenitor	6,000	□

11.2 Leaderboard

The Leaderboard page ranks all SteemBiota participants by XP. It fetches up to 200 creature posts via cursor-based pagination (Steem's API hard-limits responses to 100 posts per call), then fetches each author's recent comment history in parallel to count feed contributions. All five XP sources are included in the ranking. Each entry displays the user's rank badge, XP bar, and a breakdown of their activity across all five XP categories.

12. Blockchain Post Structure

All SteemBiota data is stored in the json_metadata.steembiota field of Steem posts and replies. The structures are versioned (version: "1.0") to support future protocol evolution.

12.1 Creature Posts

Founder creature (json_metadata.steembiota):

```
{   "version": "1.0",   "type": "founder",   "genome": { "GEN": 42, "SX": 0, "MOR":
1234, "APP": 5678,              "ORN": 9012, "CLR": 180, "LIF": 100,
"FRT_START": 30, "FRT_END": 70, "MUT": 1 },   "name": "Vyrex Nymwhisper",
"genusName": "Vyrex",   "age": 0,   "lifecycleStage": "Baby" }
```

Offspring — additional fields only:

```
{   "type": "offspring",   "parentA": { "author": "alice", "permlink": "vyrex-
nymwhisper-..." },   "parentB": { "author": "bob",   "permlink": "vyrex-shadowpaw-
..." },   "mutated": false,   "speciated": false }
```

12.2 Activity Replies

Feed reply:

```
{ "version": "1.0", "type": "feed",   "creature": { "author": "alice", "permlink":
"vyrex-nymwhisper-..." },   "feeder": "carol", "food": "nectar", "ts": "2026-01-
03T07:00:00Z" }
```

Play/Walk replies use type: "play" (field: player) or type: "walk" (field: walker).

12.3 Post Tags

Each creature post is tagged with a set of discoverable Steem tags built dynamically from the creature's name and genome. Founder tags include: steembiota, gaming, evolution, plus the genus name, first name (species part), last name (genus part), and sex. Offspring add breeding to this set. Using the creature's own name and genus as tags makes every creature discoverable through Steem's existing tag and search infrastructure, allowing communities to self-organise around genera, names, and sexes.

12.4 Permlinks

Post permlinks are derived from the post title: lowercased, whitespace replaced with hyphens, non-alphanumeric characters stripped, truncated at 200 characters, and a millisecond timestamp appended. This guarantees uniqueness across all posts from the same author.

13. Security Considerations

13.1 No Private Keys

SteemBiota never handles private keys. All blockchain writes are delegated to Steem Keychain, a browser extension that signs transactions locally. The application receives only a success or failure response, never the signing key itself.

13.2 Client-Side Rule Enforcement

All anti-spam rules, kinship checks, and permit validations are enforced client-side. Because Steem has no smart contract layer, a determined bad actor could bypass these rules by writing directly to the Steem API. SteemBiota mitigates this through transparency: any non-compliant post is visible on-chain and will receive the appropriate provenance badge (Duplicate, No Parents, Unverified Origin) when viewed in any compliant client.

13.3 Phantom Ancestry Guard

The kinship checker throws an error — rather than silently passing — when a phantom ancestor is encountered in the breeding chain. This prevents actors from deleting intermediate ancestor posts to erase a kinship relationship and then using formerly-related creatures as breeding partners.

13.4 Transfer Handshake

The two-sided transfer protocol prevents griefing attacks where an actor could spam other users' profiles with unwanted creatures. A transfer offer has no effect on ownership until the named recipient explicitly publishes an on-chain acceptance.

14. Conclusion

SteemBiota demonstrates that a fully-featured interactive game ecosystem — with genetics, lifecycle management, social breeding, verified ownership, and anti-cheating provenance — can be built on top of an existing public blockchain without any backend infrastructure, centralised database, or trusted operator.

Every creature is a permanent, auditable artefact. Every interaction is a signed, timestamped blockchain transaction. The rules of the game are enforced through transparency rather than restriction: non-compliant creatures are labelled, not deleted; legitimate creatures are mathematically verifiable from first principles by any reader.

The project also illustrates a broader principle: blockchain social platforms like Steem, which store structured metadata alongside user content, are unusually well-suited as substrates for persistent digital ecosystems. The infrastructure for content discovery, author identity, and reply threading already exists — a game developer need only define the genome schema and the interaction rules, and let the blockchain handle the rest.

SteemBiota is open source. The community is encouraged to experiment, fork, and extend the ecosystem. The blockchain ensures that whatever is built on top of it will remain readable and verifiable long after any particular client application ceases to be maintained.

https://puncakbukit.github.io/steembiota

Created by @puncakbukit for the Steem blockchain ecosystem.